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SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Total Marks: 50

Code of Conduct

I ASWATHA-V - 192525390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :

1. Comparison of type -1 and type-2 Hypervisors

Introduction :-

A hypervisor is software that creates and manages virtual machines on physical hardware. It allocates CPU, memory and storage resources to support multiple workloads in a cloud datacenter.

Types of Hypervisors

* Type -1 : Runs directly on physical hardware Example
  VMware ESXi

* Type -2 : Runs on a host operating system. Example
  Oracle Virtual Box.

Comparison

Feature	Type -1	Type - 2
Architecture	Directly on hardware	on host OS
performance	High	comparatively lower
Security	smaller attack surface	peperds on host OS
management	centralized	mainly individual system
Usage	cloud datacenters	Testing and development

performance security and manageability

* Type-1 provides efficient hardware access with less overhead.
* Type-2 requires additional resources for the host operating system.
* Type-B reduces dependence on a general-purpose host OS.
* Both require security updates and proper VM isolation.
* Type-1 supports centralized management and workload migration
* Type-2 is easier to install for personal testing and development.

Benefits:-

* improves hardware Utilization
* Supports multiple operating systems.
* Enables flexible resource allocation.
* Reduces the need for separate physical servers.

conclusion:

Type-1 hypervisors are better for mission critical cloud application because they provide performance scalability, and centralized management.

2. Workload Isolation in multi-Tenant cloud Environments

Introduction:-

cloud proriders host finance, healthcare, and student portals on shared physical hardware. virtualizabion separates their workloads and protects sensitive information from unauthorized access.

Virtualization:- Based Isolation:-

* VM isolation: Rus each workload in a Sparate virtual machine.
* Network Isolation: Uses virtual networks. VLANs, and firewalls to restrict Communication.
* storage isolation: Uses Seperte virtual disks and access permissions.
* Resource isolation: Allocates CPU and memory to prorevent resource competition.
* Access control: Restricts sensitive resources to authorized users.

TWO Major Attack surfaces

1. Hyperrisor vulnerabilites
   * Attackers may exploit hyperrisor flaws to escape a VM.
   * A successful attack may affect the host or other VMs.
   * Regular patching and strong isolation reduce this risk

2. Management interfaces

- Weak passwords and exposed APIs may allow unauthorized access.
- Incorrect firewall rules may expose sensitive workloads
- Multi-factor authentication and restricted access improve security.

Limitations

- Incorrect configuration can weaken isolation.
- Hypervisor vulnerabilities may affect multiple VMs.
- Shared resources may cause performance issues.

Security measures

- Apply regular security updates.
- Encrypt sensitive data.
- Use network segmentation and firewalls
- Monitor access logs and suspicious activities.
- Enforce least-privilege access.

Conclusion

Strong VM isolation, network segmentation, encryption, and continuous monitoring help protect finance healthcare, and student portal workloads in shared cloud environments.

3. VM Host Utilization Analysis and Corrective Action.

Introduction:

VM host utilization measures how efficiently physical server resources support virtual machines. High utilization can cause slow application responses and delayed VM start up.

Analysis of Given values

Metric	value	possible Issue
CPU	92%	High processing load
Memory	88%	Limited free memory
storage latency	35ms	Slow Storage operations
VM boot time	170 seconds	Delayed Startup

Likely Bottlenecks

- ★ High CPU usage may delay application processing
- ★ High memory usage may cause memory pressure or swapping
- ★ Storage latency may indicate overloaded disks or I/O queues.
- ★ Slow Storage operation may increase VM startup time
- ★ Simultaneous VM startups may increase resource contention.

Corrective Action

- Migrate VM to less-loaded hosts.
- increase CPU or memory capacity when necessary.
- Use faster SSD or NVMe storage.
- stagger VM startup service operations.
- Monitor resource utilization continuously.
- Optimize unnecessary startup services.

Benefits.

- Improves application response time.
- Reduces VM boot delays.
- prevents resource exhaustion.
- Improves server efficiency.

Limitations

- Hardware upgrades increase costs.
- Migration requires sufficient host capacity
- Excessive VM consolidation may create new bottlenecks.

Conclusion

High cpu usage and memory pressure, combined with storage lantency, may be affecting VM performance load balancing resource optimization help resolve identity mismatches and secure access multiple cloud.

4. Software - Defined Datacenter (SDDC)

Introduction :-

A software - Defined Datacenter (SDDC) uses software to manage and automate compute, storage, and network resources. It improves flexibility and reduces manual work.

Major components

* Compute virtualization : creates and manages virtual machines.
* Storage virtualization : combines storage resources into logical pools.
* network virtualization : creates virtual networks and applies security policies
* centeralized management : Controls resources though software tools.

Comparison

Feature	Traditional datacentor	SDDC
provisioning	manual	Automated
scaling	Hardware-dependent	flexiable
Management	separate tools	centralized
Resource usage	Less flexible	Efficient

Benefits

- Enables faster application deployment
- Improves hardware utilization
- Reduces manual administration.
- Supports flexible scaling.

Limitation:

- Initial setup can be expensive
- Requires skilled administrators
- Incorrect configurations may create security risks.

Conclusion:-

SDDC improves agility through automation and virtualization. It provides faster provisioning flexible scaling and efficient resource management.

5. Multi-cloud federation: Identity and privacy

Introduction

Multi-cloud federation allows users to access service across different cloud providers through trusted indentity system. Identity mismatches occur when providers use different authentication methods and permission structures.

causes of Identity mismatches

* Different User IDs and identity formats.
* Incompatible authentication protocols.
* Different roles & permission

Security Design:

* Federated identity
* Standard protocols
* Role Mapping
* secure tokens
* Privacy protection
* Monitoring

Benefits:

* simplifies access
* Reduces the need for separate credentials
* Improves authentication & access control
* Protects sensitive information.

Limitation :

* Federation setup can be complex
* Identity provider failures may affect services
* Incorrect role mapping.

Conclusion :

Secure federation uses standard protocols, consistent role mapping & strong authentication to resolve identity mismatches & protect privacy across multiple cloud providers.